

Indian Institute of Technology, Gandhinagar
ES226 Semiconductor Devices
Semester 2, Tutorial – 8

Q1. How can diode be used to make logic gates (AND, OR, NAND, NOT, NOR) whichever possible?

Q2. How to make read only memory using diodes?

Q3. Which region can be used for voltage regulation, rectifier, clipping/clamping circuits using diode? Plot and explain the operation using suitable input and circuit configuration.

Q4. Explain the photodetector application of the diode?

- a. Plot and Explain the circuit operation briefly and which region of the diode I-V is used for the same?
- b. What is photodetector gain?
- c. Derive and explain the different parameter dependencies.
- d. Explain different methods to increase the photodetector efficiency?

Q5. How to make a solar cell using diode?

- a. Plot and Explain the circuit operation briefly and which region of the diode I-V is used for the same?
- b. Describe when maximum power is delivered?
- c. Explain figure of merits such as efficiency, fill factor and the parameter they depend on.
- d. Explain different methods to increase the solar cell efficiency?
- e. Why Si is used even if efficiency is not so good?

Q6. Describe the LED and LASER operation qualitatively?

- a. Which region of diode IV is used for LED operation?
- b. What are the manufacturing dependencies one needs to take care for efficient light illumination?
- c. How can the LED efficiency be improved?

Q7. Explain the operation of MOS Capacitor with P- type semiconductor and sketch the energy band diagram for each mode operation.

- a) Accumulation mode
- b) Depletion mode
- c) Strong Inversion mode

Q8. For the MOSCAP with P-Type substrate.

- 1) Plot the graphs for surface potential versus gate voltage
 - 1) Oxide thickness variation

- 2) Channel doping variation
- 2) Why surface potential saturates after threshold voltage? Give the proper reason.

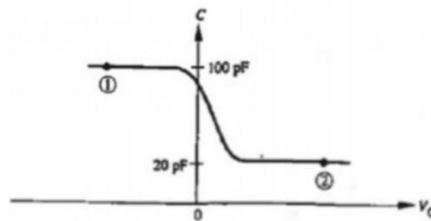
Q9. For the MOSCAP with P-Type substrate.

- Plot the graphs for silicon charge versus surface potential
- Plot the graphs for silicon charge versus gate voltage

Q10. For the MOSCAP with P-Type substrate.

- Plot the C-V characteristics of the HFCV, LFCV and Deep Depletion CV, Explain the behaviour and give the proper reason.

Q11. The CV characteristics exhibited by a MOSCAP is displayed in the figure below:



- Are the semiconductor components of MOSCAP n-type or p-type?
- If the area of MOS-C is $3 \times 10^{-3} \text{ cm}^2$ what is the thickness of oxide?
- Determine the depletion width.

Q12. A MOS capacitor consists of a p-type silicon substrate and an n^+ poly-silicon gate. The oxide thickness is $22 \text{ \AA}$ (2.2 nm), the substrate doping concentration is $N_A = 3 \times 10^{17} \text{ cm}^{-3}$, and the gate doping concentration is $N_D = 1 \times 10^{20} \text{ cm}^{-3}$. The intrinsic carrier concentration is $n_i = 1.45 \times 10^{10} \text{ cm}^{-3}$. The relative permittivity of silicon and silicon dioxide are 11.7 and 3.9, respectively. The oxide layer contains 2×10^{10} positive ions per cm^2 (i.e., $Q_{ox} = 2 \times 10^{10} \times q$, where $q = 1.6 \times 10^{-19} \text{ C}$).

Using the given data, calculate the following parameters for the MOS capacitor:

- The oxide capacitance per unit area.
- The maximum depletion width in silicon.
- The maximum electric field in silicon and in the oxide.
- The flat-band voltage.
- The threshold voltage.

Q13. A PN junction is used as a photodetector.

- Explain why a simple PN junction photodiode has limited absorption efficiency.
- How does a PIN photodiode improve absorption efficiency? What is the trade-off of

increasing the intrinsic region thickness?

(c) Explain the working principle of an Avalanche Photodiode (APD) using impact ionization.

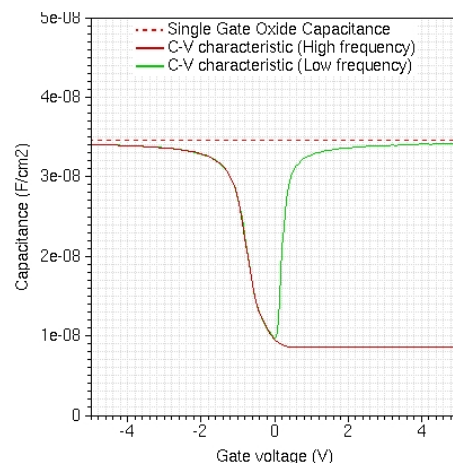
Q14. Calculate the threshold voltage of a silicon nMOS capacitor with a substrate doping $N_a = 10^{17} \text{ cm}^{-3}$, a 20 nm thick oxide ($\epsilon_{ox} = 3.9 \epsilon_0$) and an aluminum gate ($\Phi_M = 4.1 \text{ V}$). Assume there is no fixed charge in the oxide or at the oxide-silicon interface.

Q15. Calculate the oxide capacitance, the flatband capacitance and the high frequency capacitance in inversion of a silicon nMOS capacitor with a substrate doping $N_a = 10^{17} \text{ cm}^{-3}$, a 20 nm thick oxide ($\epsilon_{ox} = 3.9 \epsilon_0$) and an aluminum gate ($\Phi_M = 4.1 \text{ V}$).

Q16. Plot Band Diagram, Potential, E field, charge density, carrier concentration for MOSCAP in accumulation, depletion and inversion region.

Q17. For a MOSCAP with SiO_2 as oxide and Si as semiconductor, a CV measurement is shown. Answer following:

- Calculate the Oxide thickness.
- What is the type of the semiconductor?
- Calculate the doping of the semiconductor.
- Calculate the Threshold voltage V_{th} (Assume work function of metal and semiconductor are equal i.e. $\phi_m = \phi_{Si}$)
- Draw the band-diagram in equilibrium and inversion & mark the values. (Assume $\phi_m = \phi_{Si}$)
- Find the midgap voltage and midgap capacitance
(Midgap voltage is the gate voltage at which the doped Si behaves as if intrinsic exactly at the oxide/silicon interface)



if $\phi_m \neq \phi_{Si}$

- Calculate the value of Flatband capacitance, Flatband Voltage using the same CV measurements.

- b. Calculate the actual threshold voltage.

Q18. Derive the sub-threshold slope for MOSFET. What could be an ideal subthreshold slope for the MOSFET? Can this ideal SS be achieved? What are the limitations and challenges?